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SMA 2101: Calculus I

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and are accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 10

More Revision Examples

Example  Find the points on $y = x^4 + 4x^3 - 6$ at which the gradient is zero and determine the nature of the points

Solution: $\frac{dy}{dx} = 4x^3 + 12x^2$ $\frac{d^2y}{dx^2} = 12x^2 + 24x$

At the stationery point, $\frac{dy}{dx} = 0$ therefore

$$4x^3 + 12x^2 = 0 \quad \text{or} \quad 4x^2(x + 3) = 0$$

$$\Rightarrow x = -3 \quad \text{or} \quad x = 0$$

When $x = -3$, $\frac{d^2y}{dx^2} = 12(-3)^2 + 24(-3) = 36 > 0$

therefore y has a minimum value at $x = -3$

$$y = (-3)^4 + 4(-3)^3 - 6 = -33$$

Thus $(-3, -33)$ is a minimum turning point

When $x = 0$, $\frac{d^2y}{dx^2} = 12(0)^2 + 24(0) = 0$





therefore y has a an inflection at $x = 0$

$$y = (0)^4 + 4(0)^3 - 6 = -6$$

Thus $(0, -6)$ is a point of inflection

□

Example . Find the maximum and minimum points of the curve $y = x^2(x + 1) = x^3 + x^2$

Solution: $\frac{dy}{dx} = 3x^2 + 2x$ $\frac{d^2y}{dx^2} = 6x + 2$

At the stationery point, $\frac{dy}{dx} = 0$ therefore

$$3x^2 + 2x = 0 \quad \text{or} \quad x(3x + 2) = 0$$

$$\Rightarrow x = -\frac{2}{3} \quad \text{or} \quad x = 0$$

$$\text{At } x = -\frac{2}{3}, \quad \frac{d^2y}{dx^2} = 6\left(-\frac{2}{3}\right) + 2 = -2 < 0 \implies \text{maximum}$$

value

$$y = \left(-\frac{2}{3}\right)^3 + \left(-\frac{2}{3}\right)^2 = \frac{4}{27}$$





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$$\text{At } x = 0, \quad \frac{d^2y}{dx^2} = 6(0) + 2 = 2 > 0 \implies \text{minimum value}$$

$$y = (0)^3 + (0)^2 = 0$$

□

Example . Find the minimum and maximum value of the function $2 \sin t + \cos 2t = 2 \sin t + \cos^2 t - \sin^2 t$

Solution: $\frac{dy}{dt} = 2 \cos t - 2 \sin t \cos t - 2 \sin t \cos t = 2 \cos t (1 - 4 \sin t)$

$$\frac{dy}{dt} = 0 \text{ for maximum or minimum}$$

$$\frac{dy}{dt} = 2 \cos t (1 - 4 \sin t) = 0$$

$$\implies \cos t = 0 \text{ which gives } t = \frac{\pi}{2} \text{ or}$$

$$\implies 1 - 2 \sin t = 0 \text{ which gives } \sin t = \frac{1}{2}, t = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\frac{d^2y}{dt^2} = -2 \sin t (1 - 4 \sin t) + 2 \cos t (-4 \cos t) = -2 \sin t + 8 (\sin^2 t - \cos^2 t)$$

$$-2 \sin t - 8 \cos 2t$$





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$$\text{at } x = \frac{\pi}{6}, \quad \frac{d^2y}{dt^2} = -2 \sin\left(\frac{\pi}{6}\right) - 8 \cos\left(\frac{\pi}{3}\right) = -1 - 4 = -5 < 0$$

maximum point

$$\Rightarrow \text{maximum value; } y = 1 + \frac{1}{2} = \frac{3}{2}$$

$$\text{at } x = \frac{\pi}{2}, \quad \frac{d^2y}{dt^2} = -2 \sin\left(\frac{\pi}{2}\right) - 8 \cos(\pi) = -2 + 8 = 6 > 0$$

minimum point

$$\Rightarrow \text{maximum value; } y = 2 - 1 = 1$$

□

Example  Find the turning points and the point of inflection of the curve $y = x^5 - 5x^4 + 5x^3 - 1$

$$\text{Solution: } \frac{dy}{dx} = 5x^4 - 20x^3 + 15x^2 \qquad \frac{d^2y}{dx^2} = 20x^3 - 60x^2 + 30x$$

At the stationery point, $\frac{dy}{dx} = 0$ therefore

$$5x^4 - 20x^3 + 15x^2 = 0 \quad \text{or} \quad 5x^2(x^2 - 4x + 3) = 0 \quad \text{or} \quad 5x^2(x - 3)(x - 1)$$



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$$\Rightarrow x = 3 \quad \text{or} \quad x = 0 \quad \text{or} \quad x = 1$$

$$\text{At } x = 3, \quad \frac{d^2y}{dx^2} = 20(3)^3 - 60(3)^2 + 30(3) = 90 > 0 \Rightarrow \text{minimum value}$$

$$y = (3)^5 - 5(3)^4 + 5(3)^3 - 1 = -28, \text{ the minimum point is } (3, -28)$$

$$\text{At } x = 0, \quad \frac{d^2y}{dx^2} = 20(0)^3 - 60(0)^2 + 30(0) = 0 \Rightarrow \text{point of inflection}$$

$$y = (0)^5 - 5(0)^4 + 5(-)^3 - 1 = -1, \text{ the point of inflection is } (0, -1)$$

$$\text{At } x = 1, \quad \frac{d^2y}{dx^2} = 20(1)^3 - 60(1)^2 + 30(1) = -10 < 0 \Rightarrow \text{maximum value}$$

$$y = (1)^5 - 5(1)^4 + 5(1)^3 - 1 = 0, \text{ the maximum point is } (1, 0)$$

□





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EXERCISE 1.  (1) $y = e^{e^x}$

(2) $y = xe^y + y = 4$

(3) $y \ln x = xe^y$

(4) $(1 + e^{3x})^2 = 3 + \ln(x + y)$

(5) Find $\frac{dy}{dx}$ at $(1, 2)$ if (i) $x + xy + y^2 = 7$

(ii) $x^3 = (y - x^2)^2$

(6) If $xe^y + \sin xy + y - \ln 2 = 0$,

find $\frac{dy}{dx}$ at point $(0, \ln 2)$

10.0.1. More Revision Questions

1. Find the values at the point inflexion (if any) of the curve

$$y = 3x^4 - 4x^3 + 1$$

2. Discuss the nature of points on the curve
- $y = 3x^4 - 8x^3 - 24x^2 + 96x$
- at the point which the tangent to the curve is





parallel to the x -axis

3. Find the nature of the stationary points of the function

$$y = 3x^5 + 6x^4 - 4x^2 + 1$$

4. Evaluate $\lim_{h \rightarrow 0} \left(\frac{2^x - 3^x}{x} \right)$

5. Find y' if

(a) $y = 3^x$

(b) $y = \cos(\sin^3 3x)$

(c) $y = \sqrt{x + \sqrt{3x + \sqrt{2x}}}$

(d) $y = \cos^{-1} \left(\frac{x-1}{x^2+1} \right)$

(e) $(x+y)^3 + (x-y)^3 = x^4 + y^4$

(f) $y =$

$x \sin^{-1}(3x - \sqrt{1 - 9x^2})$

(g) $y = \ln(\cot x - \csc x)$

(h) $y =$

$x^2 \ln(\sin^{-1}(x^2 - 1))$

(i) $y = \log_{10}(x^2 + 4x) + x^x \sin(x^2 + 1)$





(j) $y = x^{-x} \log_{10}(x^2 + 1) - x^3 \cos^{-1}(5x)$

(k) $y = e^{x \sin^{-1}(x^2)} + \ln(\tan 2x) + \log_{10} e^{x^2} + x$

6. If $y = \left(\frac{1+\sin x}{1-\sin x}\right)^{1/2}$ show that $\frac{dy}{dx} = \frac{1}{1-\sin x}$



Solutions to Exercises

Exercise 1. (1) $e^x e^{e^x}$

(3) $\frac{xe^y - y}{x(\ln x - xe^y)}$

(5) (i) $\frac{-3}{5}$

(6) (-2.6932)

(2) $\frac{e^y}{xe^y + 1}$

(4) $6e^{3x} (1 + e^{3x}) (x + y) - 1$

(ii) $\frac{7}{2}$

Exercise 1